OPEN SCIENCE IN SPORT: IT'S THAT IMPORTANT

Garrett Bullock, PT, DPT, DPhil

LITERATURE REVIEW

GARRETT S. BULLOCK, PT, DPT, DPhil[1,2,3,4] • PATRICK WARD, PhD[5] • FRANCO M. IMPELLIZZERI, PhD[6]
STEFAN KLUZEK, FRCP, FFSEM, DPhil[3,7,8] • TOM HUGHES, PT, PhD[9] • CHARLES HILLMAN, MBChB[7] • BRIAN R. WATERMAN, MD[1]
KERRY DANELSON, PhD[1] • KAITLIN HENRY, MSc[1] • EMILY BARR, BS[1] • KELSEY HEALY, BS[1] • ANU M. RÄISÄNEN, PT, PhD[10,11]
CHRISTINA GOMEZ, PT, DPT, SCS, OCS, FAOOMPT[10] • GARRETT FERNANDEZ, BS[1] • JAKOB WOLF, BS[1] • KRISTEN F. NICHOLSON, PhD[1]
TIM SELL, PT, PhD[12] • RYAN ZEREGA, MS[12] • PAULA DHIMAN, PhD[13] • RICHARD D. RILEY, PhD[14] • GARY S. COLLINS, PhD[13]

Up Front and Open? Shrouded in Secrecy? Or Somewhere in Between? A Meta-Research Systematic Review of Open Science Practices in Sport Medicine Research

Sports Medicine
https://doi.org/10.1007/s40279-023-01849-6

CURRENT OPINION

Check for updates

The Trade Secret Taboo: Open Science Methods are Required to Improve Prediction Models in Sports Medicine and Performance

Garrett S. Bullock[1,2,3] · Patrick Ward[4] · Franco M. Impellizzeri[5] · Stefan Kluzek[3,6,7] · Tom Hughes[8,9] · Paula Dhiman[10,11] · Richard D. Riley[12] · Gary S. Collins[10,11]

Accepted: 25 March 2023

Editorial

Call for open science in sports medicine

Garrett S Bullock [1,2] Patrick Ward,[3] Scott Peters,[4]
Amelia Joanna Hanford Arundale [5,6] Andrew Murray [7,8]
Franco M Impellizzeri,[9] Stefan Kluzek[2,10]

A NEED FOR GREATER INTERNATIONAL COLLABORATION AND TRANSPARENCY

AGGREGATED CLINICAL FINDINGS HAVE ADDED SOME VALUE BUT FURTHER PROGRESS IS NEEDED

these scientific fields do not have to contend with protecting patient medical information.[2 3] Recent requirements by the National Institute of Health and Wellcome Trust for researchers to provide open data demonstrate the importance and applicability in medical science,[2 3] with specific strategies enacted to create open and accessible medical data in both private industry and academic institutions (table 1).[2–4]

Editorial

Paving the way for greater open science in sports and exercise medicine: navigating the barriers to adopting open and accessible data practices

Garrett S Bullock [1,2] Patrick Ward,[3] Stefan Kluzek,[4]
Tom Hughes [5] Ellen Shanley [6,7]
Amelia Joanna Hanford Arundale [8] Craig Ranson,[9]
Sophia Nimphius [10] Richard D Riley [11] Gary S Collins [12]
Franco M Impellizzeri[13]

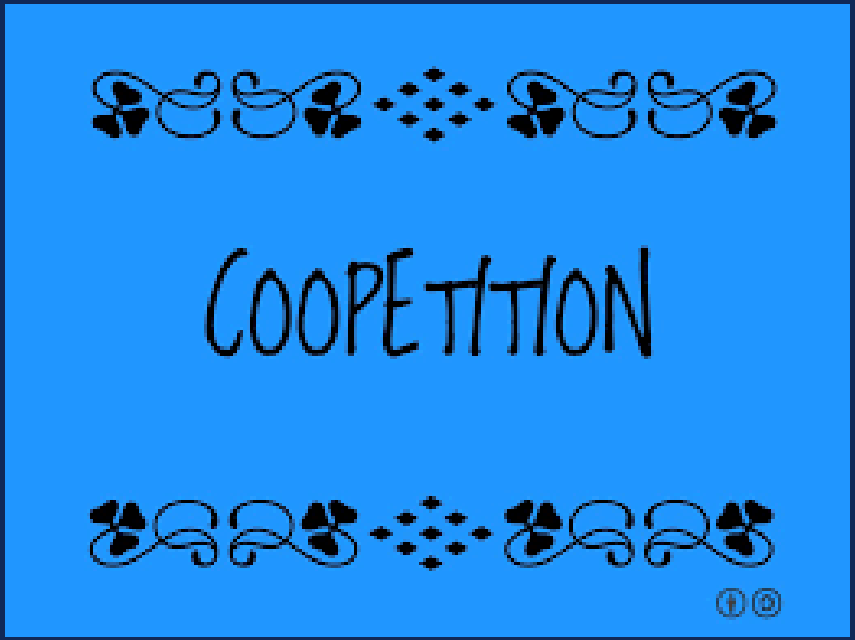
deficiencies in its implementation. To highlight this, in a study of journal editors, 71% supported open science and 78% reported they could integrate open science into their journal. However, 74% said it was not a priority, while 65% said they would not receive recognition for using open science guidelines, with daily journal activities taking higher priority.[8]

ORGANISATIONAL BARRIERS TO OPEN AND ACCESSIBLE DATA

Data sharing takes a commitment from the sporting scientific, university, leagues and governing body communities and grant funding opportunities to make this feasible, useful and actionable. There is little training for practitioners or faculty in


Cooperative


information
intellect
computer
intellect
internet
browse
Data
browse
internet
intellect
big data
Data
data storage
analyze
knowledge
knowledge
computer
internet
internet
information
computer
big data
analyze
Data

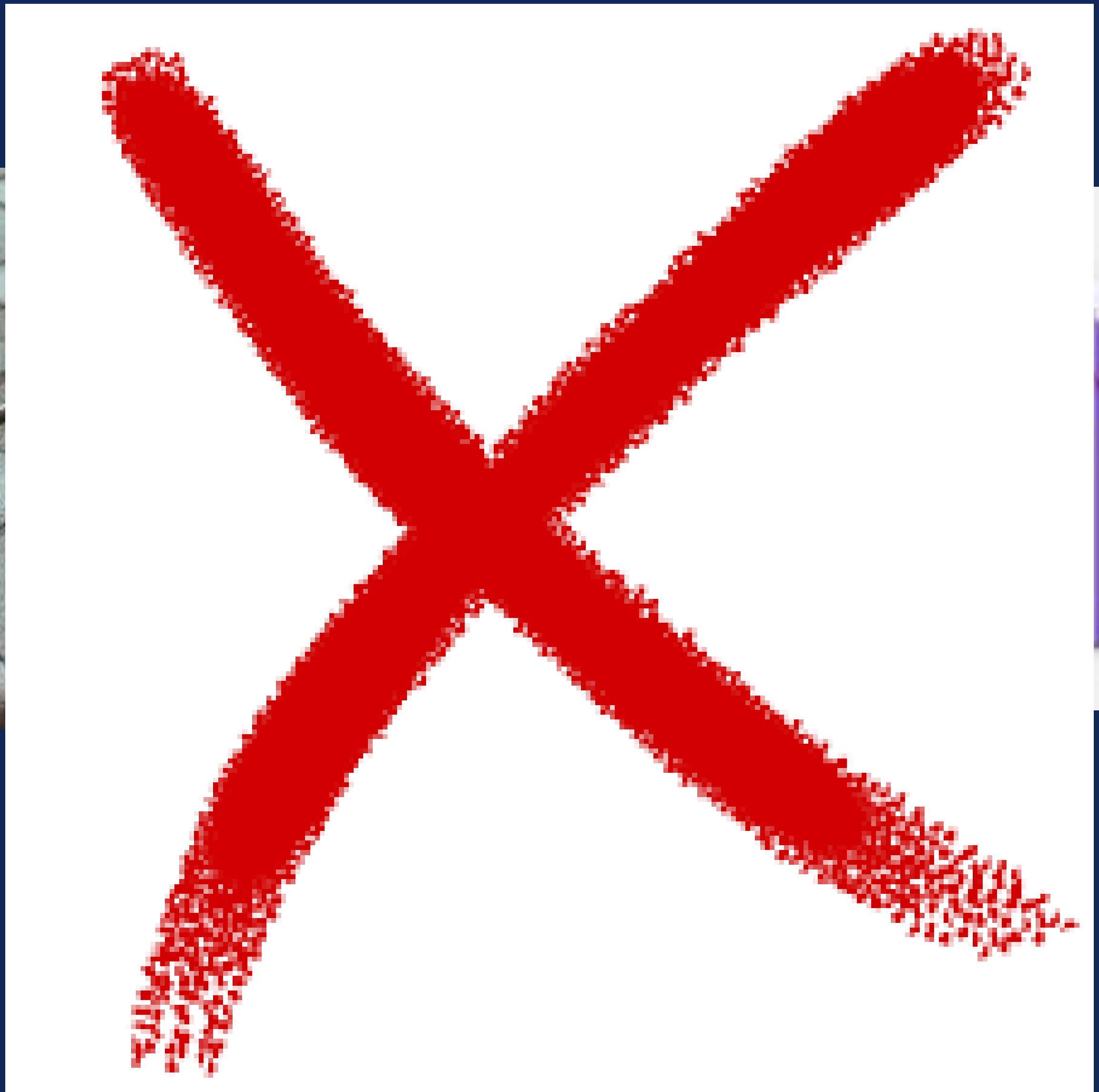

COOPETITION

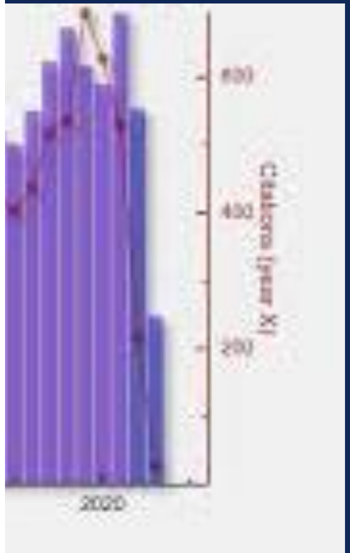

Google

So What?

Current Sport Environment = Small & Isolated

Current Sport Environment = Small & Isolated





[LITERATURE REVIEW]

GARRETT S. BULLOCK, PT, DPT, DPhil[1,2,3,4] • PATRICK WARD, PhD[5] • FRANCO M. IMPELLIZZERI, PhD[6]
STEFAN KLUZEK, FRCP, FFSEM, DPhil[3,7,8] • TOM HUGHES, PT, PhD[9] • CHARLES HILLMAN, MBChB[7] • BRIAN R. WATERMAN, MD[1]
KERRY DANELSON, PhD[1] • KAITLIN HENRY, MSc[1] • EMILY BARR, BS[1] • KELSEY HEALY, BS[1] • ANU M. RÄISÄNEN, PT, PhD[10,11]
CHRISTINA GOMEZ, PT, DPT, SCS, OCS, FAOOMPT[10] • GARRETT FERNANDEZ, BS[1] • JAKOB WOLF, BS[1] • KRISTEN F. NICHOLSON, PhD[1]
TIM SELL, PT, PhD[12] • RYAN ZEREGA, MS[12] • PAULA DHIMAN, PhD[13] • RICHARD D. RILEY, PhD[14] • GARY S. COLLINS, PhD[13]

Up Front and Open? Shrouded in Secrecy? Or Somewhere in Between? A Meta-Research Systematic Review of Open Science Practices in Sport Medicine Research

Sports Medicine
https://doi.org/10.1007/s40279-023-01849-6

CURRENT OPINION

Check for updates

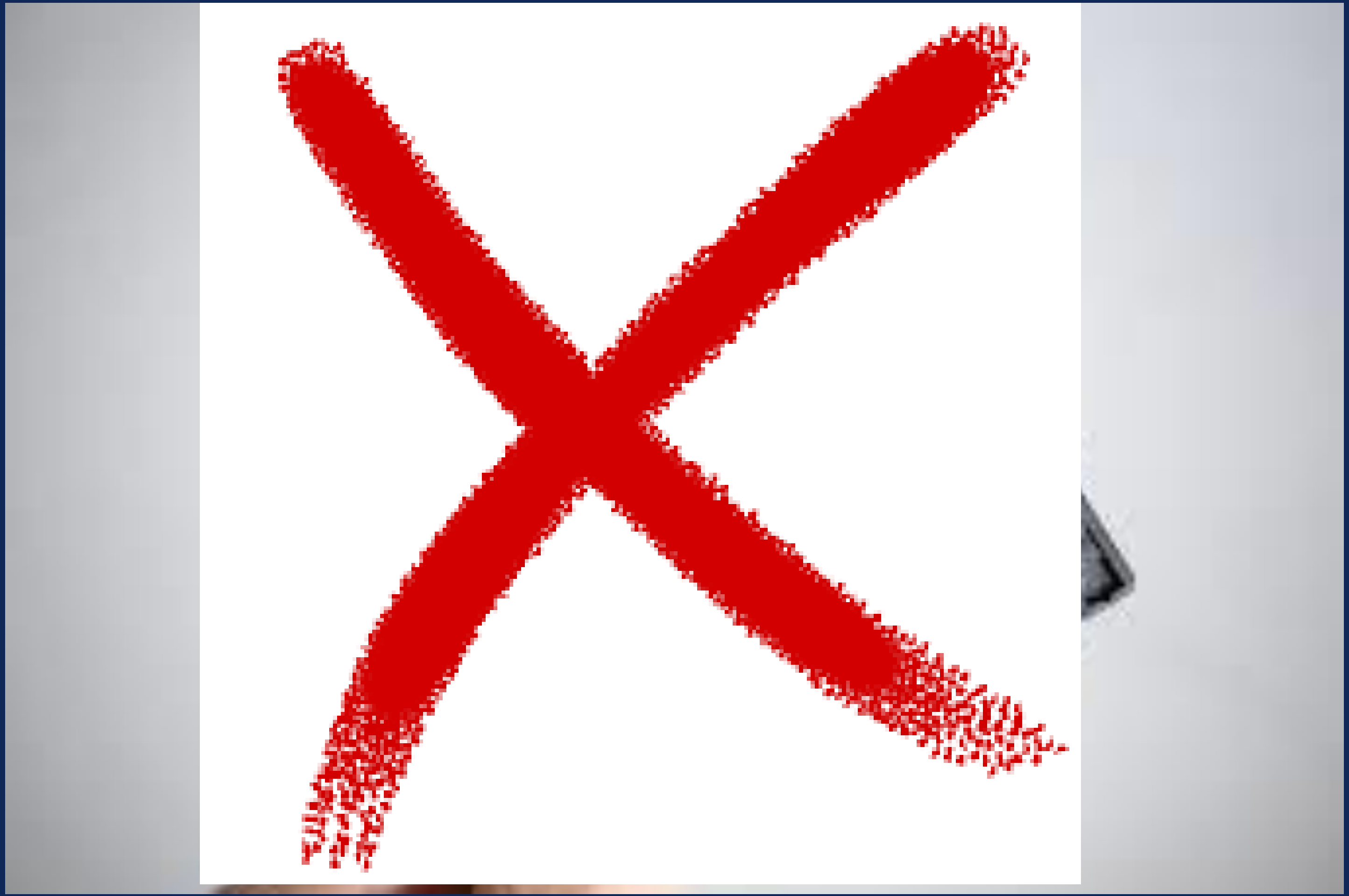
The Trade Secret Taboo: Open Science Methods are Required to Improve Prediction Models in Sports Medicine and Performance

Garrett S. Bullock[1,2,3] · Patrick Ward[4] · Franco M. Impellizzeri[5] · Stefan Kluzek[3,6,7] · Tom Hughes[8,9] · Paula Dhiman[10,11] · Richard D. Riley[12] · Gary S. Collins[10,11]

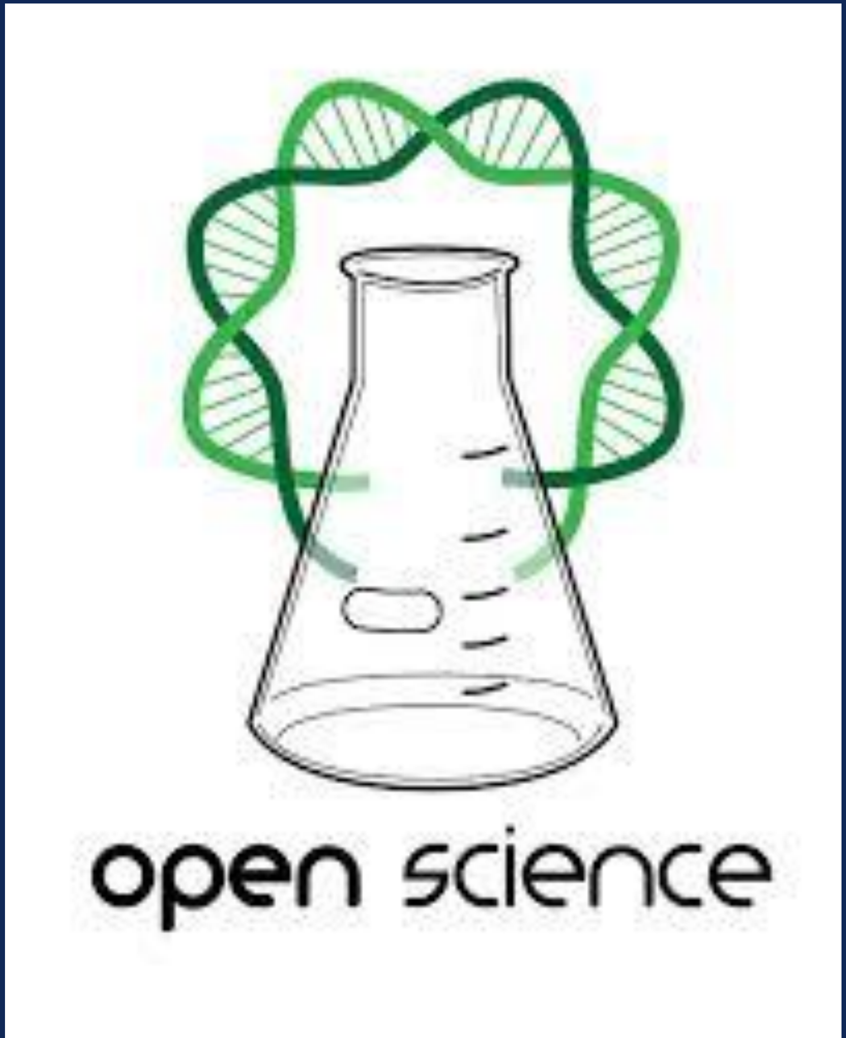
Accepted: 25 March 2023


FDA
U.S. Department of Health and Human Services
Food and Drug Administration
COVID 19
Coronavirus Vaccine

What is Open Science?



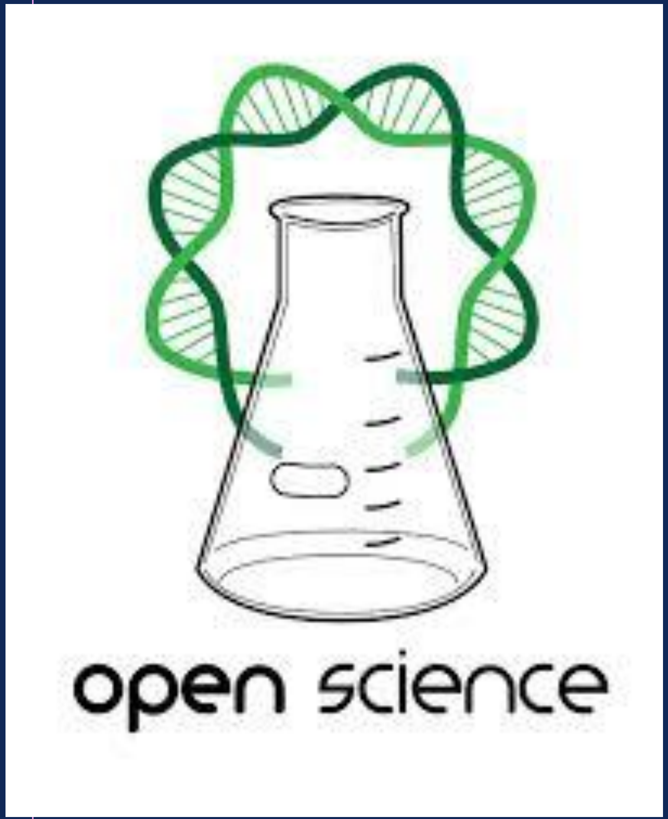
What is Open Science?



equator network

Enhancing the QUAlity and Transparency Of health Research

What is Open Science?



Findable

Accessible

Interoperable

Reusable

What is Open Science?

Findable

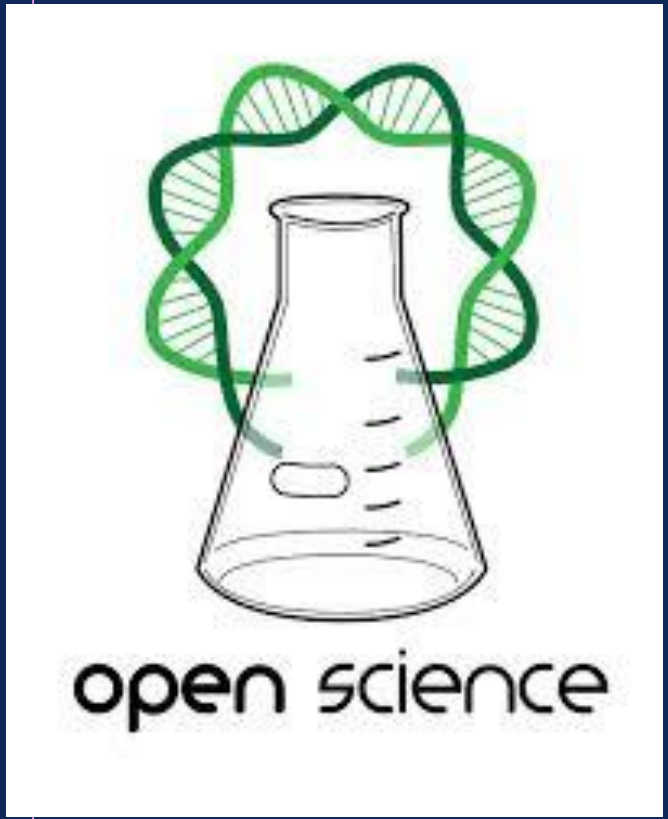
F1. (Meta)data are assigned a globally unique and persistent identifier
F2. Data are described with rich metadata
F3. Metadata clearly and explicitly include the identifier of the data they describe
F4. (Meta)data are registered or indexed in a searchable resource

What is Open Science?

Accessible

A1. (Meta)data are retrievable by their identifier using a standardised communications protocol
A1.1 The protocol is open, free, and universally implementable
A1.2 The protocol allows for an authentication and authorisation procedure, where necessary
A2. Metadata are accessible, even when the data are no longer available

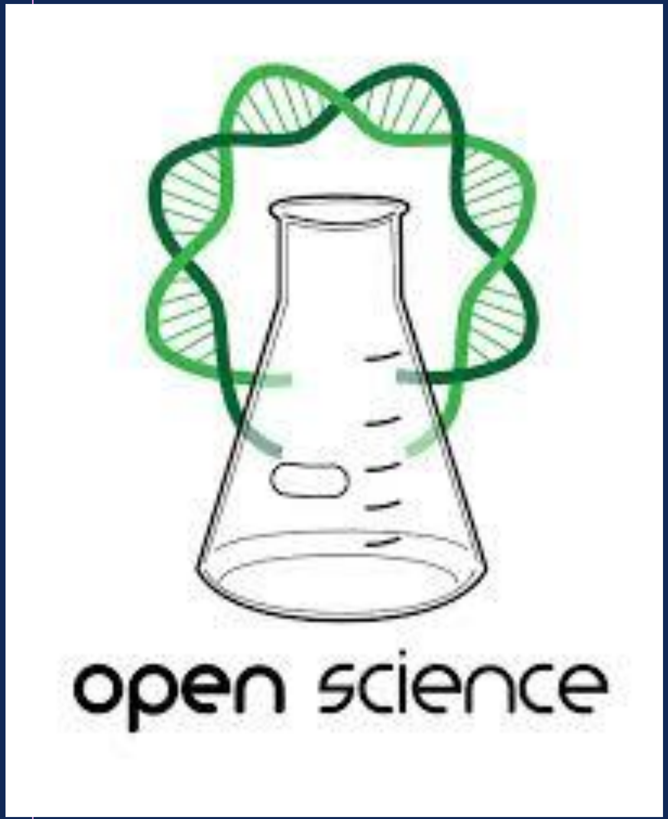
What is Open Science?



Interoperable

I1. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.
I2. (Meta)data use vocabularies that follow FAIR principles
I3. (Meta)data include qualified references to other (meta)data

What is Open Science?



Reusable

(Meta)data are richly described with a plurality of accurate and relevant attributes

R1.1. (Meta)data are released with a clear and accessible data usage license

R1.2. (Meta)data are associated with detailed provenance

R1.3. (Meta)data meet domain-relevant community standards

This is Great… But… Sport is Not Like Other Sciences


COMPETITIVE
ADVANTAGE


Identification


TRADE SECRET


DATA
GOVERNANCE

General Science Barriers for Open Science

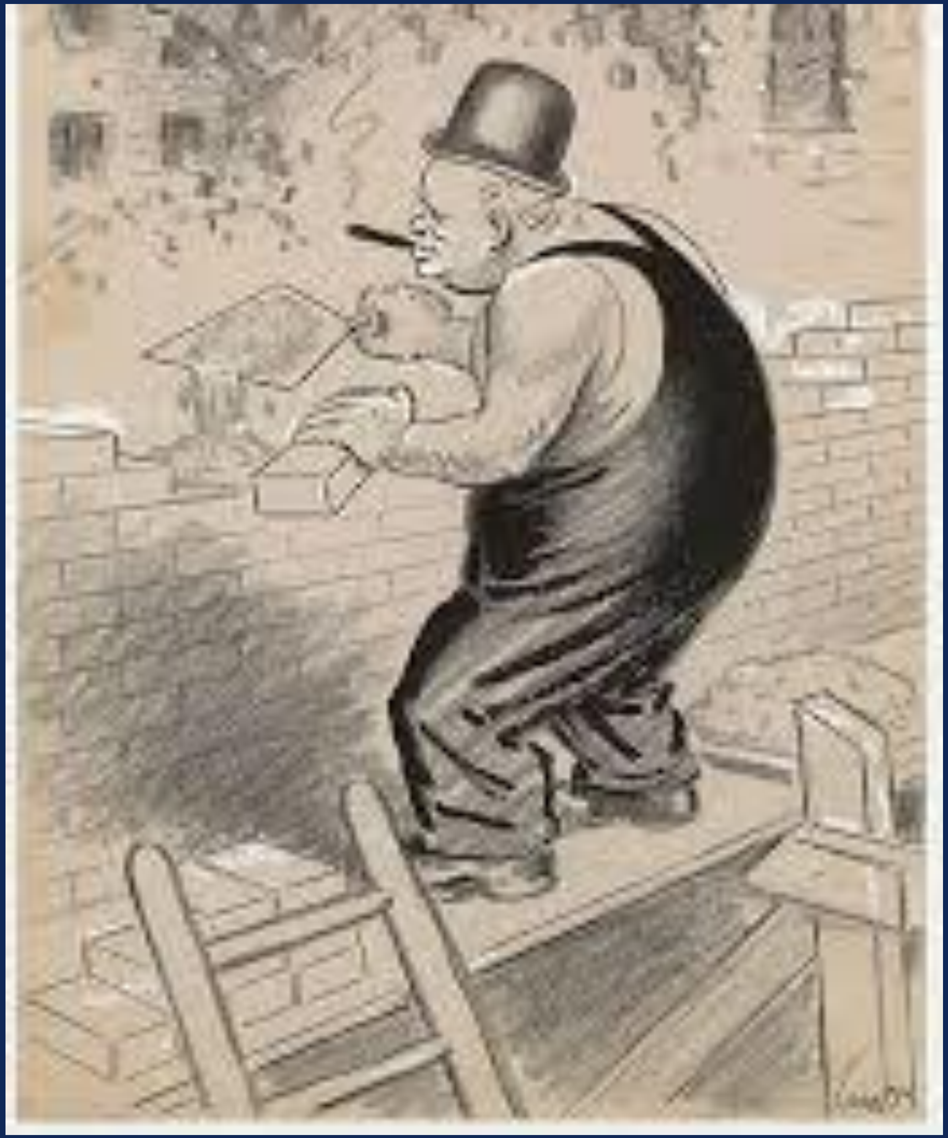
Sport Needs Unique Considerations for Open Science

De-Identification

Depersonalization

Pseudonymization

Data Caretaker

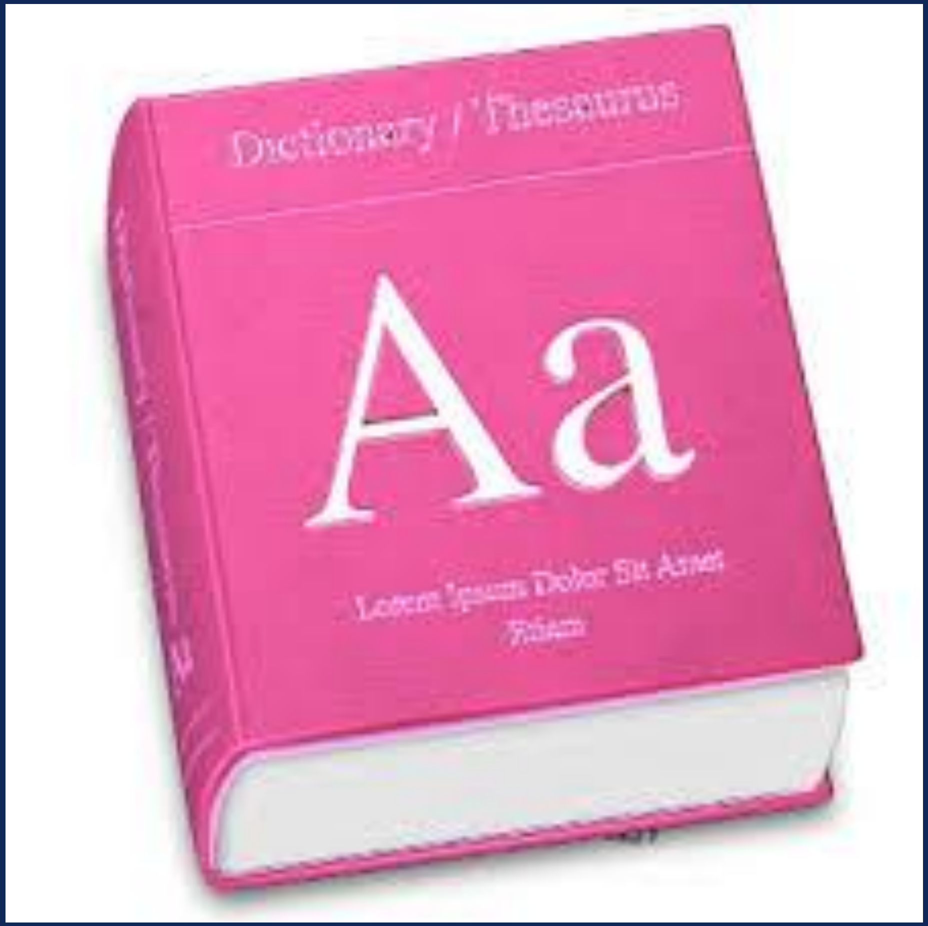


Federated Systems Warehouses

That's Great… But … What Can Individuals Do?

Educate Yourself & Others on Open Science


STANDARD
OPERATING
PROCEDURE


Dictionary / Thesaurus
Aa
Lorem Ipsum Dolor Sit Amet

Use a Standardized Strategy and System for Collection

Consensus Statement

International Olympic Committee Consensus Statement

Methods for Recording and Reporting of Epidemiological Data on Injury and Illness in Sports 2020 (Including the STROBE Extension for Sports Injury and Illness Surveillance (STROBE-SIIS))

International Olympic Committee Injury and Illness Epidemiology Consensus Group*†

Use a Standardized Strategy and System for Collection

Sport Medicine Diagnostic Coding System (SMDCS)
Orchard Sports Injury Classification System (OSICS).

Use a Standardized Strategy and System for Collection

TABLE 2
Examples: Classification of Contact as a Mechanism for Sudden-Onset Injuries[a]

Injury	Type of Contact	Example
Noncontact		
None	No evidence of disruption or perturbation of the player's movement pattern	ACL tear in a basketball player landing with knee valgus/rotation after a jump, with no contact with other players
Contact		
Indirect	Through another athlete	ACL tear in a handball player landing out of balance after being pushed on her shoulder by an opponent while in the air
Indirect	Through an object	Downhill skier suffers a concussion from a crash after being knocked off balance, hitting the gate with his knee
Contact		
Direct	With another athlete	ACL tear in a football player from a direct tackle to the anterior aspect of the knee, forcing the knee into hyperextension
Direct	With an object	Volleyball player being hit in the face by a spiked ball, resulting in a concussion

Use a Standardized Strategy and System for Collection

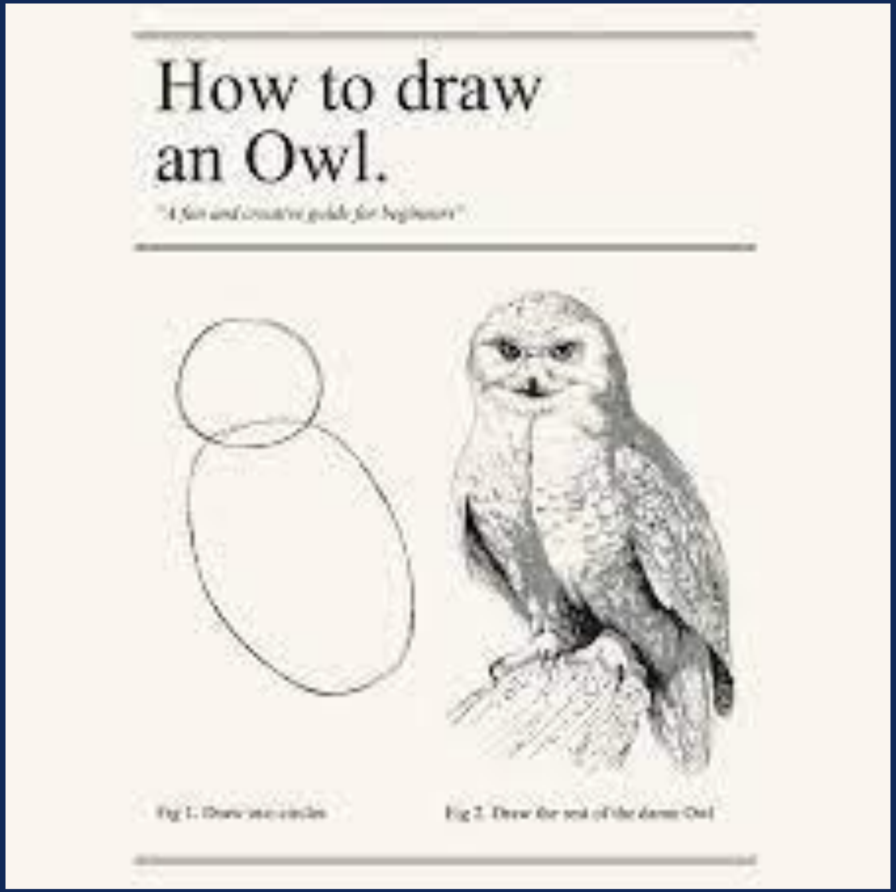
TABLE 1
Examples: Assessment of Mode of Onset

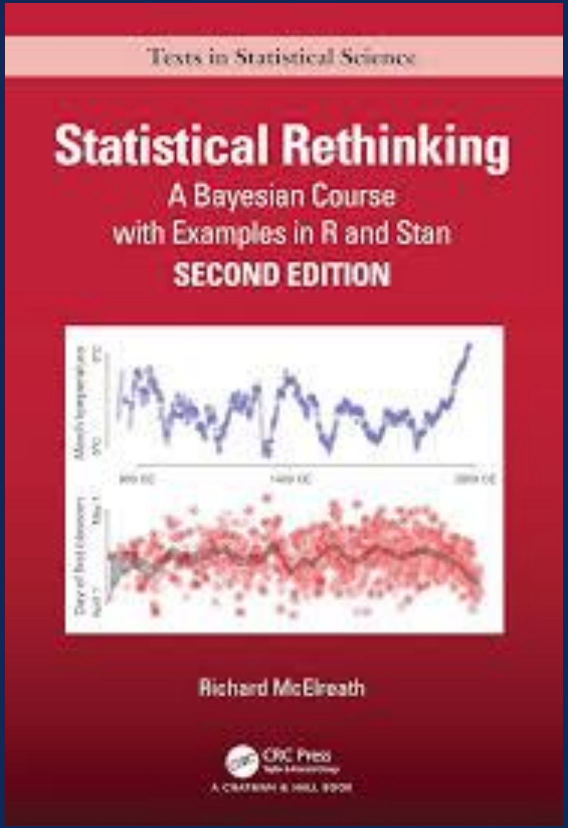
Mechanism	Presentation	Example
Acute	Sudden onset	1. A sprinter pulls up suddenly in a race, stops, and hobbles a few steps in obvious pain with a hamstring injury.
Repetitive	Sudden onset	2. A gymnast experiences a frank tibial and fibular fracture on landing from a vault; computed tomography imaging reveals pre-existing morphological changes consistent with bone stress, that is, a stress fracture.
Repetitive	Gradual onset	3. A swimmer experiences a gradual increase in shoulder pain over the course of a season; diagnosed as rotator cuff tendinopathy on magnetic resonance imaging.

Use a Standardized Strategy and System for Collection

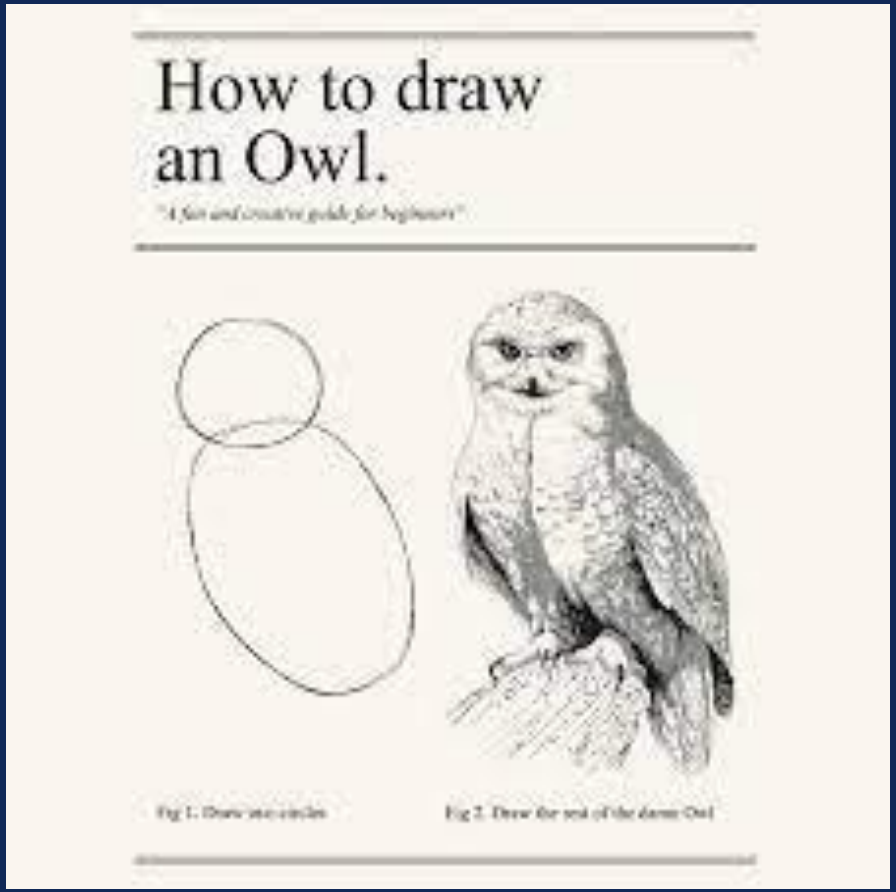
- Don’t continue to switch data collections
- Minimum ‘core’ dataset for each collection
- Decided upon as a team

DRAWING THE OWL





DRAWING THE OWL



- Understand your methods & enquiries
- Reduce error
- Replicate your scientific workflow
  - Cleaning
  - Building models
  - Analyze data
  - Inferences

Maintain Detailed Code

Maintain Detailed Code

SHARE Your Detailed Code

Open Science Takeaways

- Open science is essential for scientific advancement
- Use the FAIR Principle
- Sport is unique, but this doesn’t preclude open science practices
- Have standard operating procedures and a data dictionary
- Share Code at a minimum